Railway Fare and Reservation Calculator

1. Problem statement

Create a Python program to calculate railway fares for passengers based on distance, travel class, age-based concessions, reservation fees, and cancellation penalties.

2. Objectives

- Calculate distance-based fares.
- Apply class-wise charges.
- Provide age-based concessions.
- Calculate reservation fees and cancellation charges.

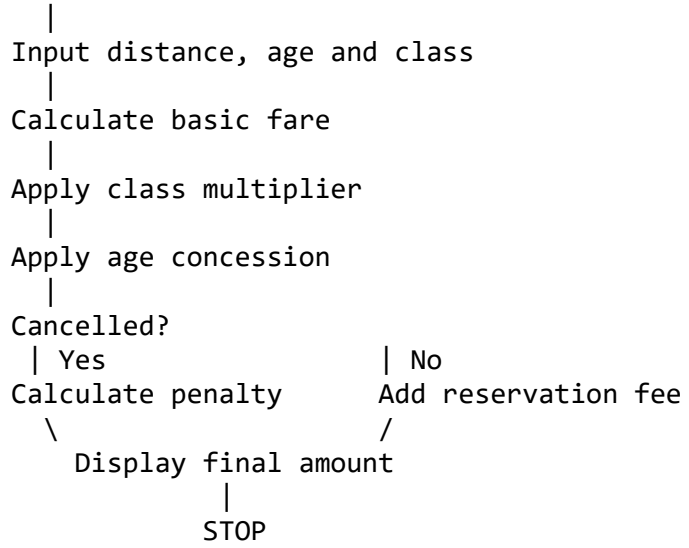
3. Assumptions

- Base fare: ₹1.50 per kilometre.
- Sleeper class multiplier: 1.0.
- AC class multiplier: 2.0.
- Children below 5 years travel free.
- Passengers aged 60 or above receive a 20% concession.
- Reservation fee: ₹40 per passenger.
- Cancellation penalty: 25% of the fare before reservation fee.

4. Algorithm

1. Input distance, passenger age, travel class, and cancellation choice.
2. Calculate the basic fare.
3. Apply the class multiplier.
4. Apply age-based concession or free travel.
5. Add reservation fee if travelling.
6. Apply cancellation penalty if cancelled.
7. Display the payable amount.

5. Flowchart



6. Pseudocode

```
START
INPUT distance, age, class, cancelled
fare = distance * 1.50
```

```
IF class = "AC"
    fare = fare * 2
END IF

IF age < 5
    fare = 0
ELSE IF age >= 60
    fare = fare * 0.80
END IF

IF cancelled
    amount = fare * 0.25
ELSE
    amount = fare + 40
END IF

PRINT amount
STOP
```

7. Python source code

```
distance = float(input("Enter distance in km: "))
age = int(input("Enter passenger age: "))
travel_class = input("Enter class (Sleeper/AC): ").lower()
cancelled = input("Cancel ticket? (yes/no): ").lower()

if distance <= 0 or age < 0:
    print("Invalid input")
elif travel_class not in ("sleeper", "ac"):
    print("Invalid travel class")
elif cancelled not in ("yes", "no"):
    print("Invalid cancellation choice")
else:
    fare = distance * 1.50

    if travel_class == "ac":
        fare *= 2

    if age < 5:
        fare = 0
    elif age >= 60:
        fare *= 0.80

    if cancelled == "yes":
        amount = fare * 0.25
        print("Cancellation penalty: Rs.", round(amount, 2))
    else:
        amount = fare + 40
```

```
        print("Total railway fare: Rs.", round(amount, 2))
```

8. Test cases

Test	Distance	Age	Class	Cancelled?	Expected amount
1	100 km	30	Sleeper	No	₹190
2	100 km	65	AC	No	₹280
3	100 km	30	AC	Yes	₹75
4	100 km	4	Sleeper	No	₹40